

CSU Education Opportunity Program: Student Success and Enrollment

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Introduction

The Educational Opportunity Program (EOP) supports students who have historically faced educational and economic barriers to college access and completion. This project examines CSU EOP enrollment and student success patterns from 2001-2024 using campus-year data from annual CSU EOP reports. The analysis asks how EOP enrollment and outcomes changed over time, how EOP students compare with the broader CSU population, how Chico State compares with systemwide patterns, and whether disruption periods such as post - 2007 and post - 2020 correspond with shifts in enrollment or outcomes.

Methodology

Data source:

The data came from annual CSU EOP enrollment Excel workbooks from 2001 - 2024. Years 2003 and 2004 were not available. The raw files contained multiple sheets, inconsistent title rows, multirow headers, repeated column names, and campus rows mixed with totals or notes. The cleaning process produced one wide campus-year dataset and one long format dataset for plotting and modeling . Due to multiple sheets and similar column names all variables were set to have the table number as a prefix for clarity (e.g. t03_freshman , t04_freshman)

The final analysis used **502 campus-year observations** across **23 CSU campuses** and **22 reporting years**.

Key Variables:

Time - year , post_2008 , post_2020 , years_after_2008 , years_after2020

Campus – campus_name , Chico indicator

Enrollment – total , Total EOP enrollment from freshman, sophomore, junior, and senior counts.

Student Success – EOP and CSU graduation and persistence rates for first time freshman (e.g. ftf_eop_grad_rate) and lower division students (e.g., t19_eop_grad) were weighted across regular FTF and exceptional admit FTF tables

Demographics – eth_ , featuring different race/ethnicity proportions over 9 variables. Older variables were aggregated to match more recent ethnicity classifications

Retention Proxy – t07_continuing_students , t07_returning_students , t07_returning_transfers , divided by total for EOP enrollment.

Feature engineering:

Missing values that reflected structural reporting changes were kept as missing, and models used the complete cases needed for each analysis.

Statistical Methods

Analyses included time series plots, campus comparisons, Wilcoxon signed rank tests, and mixed-effects models with campus as a random effect to account for repeated observations across campuses.

Sample Characteristics

The sample included **502 campus-year observations** across three time periods. Average EOP enrollment was slightly lower in the **2020+ period** compared with the earlier periods, until 2020. The demographic composition also shifted over time, with the Hispanic/Latino share increasing from **0.48 pre-2008** to **0.71 after 2020**, while White, Black, and Asian shares were lower in the later period. Student success outcomes were generally higher after 2020, especially for lower division EOP graduation and persistence rates, although overall CSU rates remained slightly higher than EOP rates for most comparable outcomes.

Table 1. Campus-year sample characteristics by period.				
Characteristic	Overall N = 502 [†]	2008-2019 N = 276 [†]	2020+ N = 115 [†]	Pre-2008 N = 111 [†]
Total EOP enrollment	1,322.08 (923.75)	1,331.42 (930.34)	1,262.25 (783.83)	1,360.84 (1,039.44)
Retention proxy	0.74 (0.10)	0.74 (0.11)	0.72 (0.09)	0.75 (0.11)
Female	0.66 (0.09)	0.66 (0.10)	0.67 (0.09)	0.66 (0.08)
Male	0.34 (0.09)	0.34 (0.10)	0.33 (0.09)	0.34 (0.08)
Hispanic/Latino	0.59 (0.16)	0.59 (0.16)	0.71 (0.11)	0.48 (0.14)
White	0.07 (0.07)	0.07 (0.07)	0.05 (0.03)	0.10 (0.08)
Black	0.09 (0.06)	0.09 (0.06)	0.06 (0.03)	0.13 (0.08)
Asian	0.12 (0.11)	0.11 (0.10)	0.10 (0.08)	0.18 (0.14)
FTF EOP graduation rate	47.56 (12.83)	48.06 (12.03)	57.27 (10.08)	36.79 (8.08)
FTF EOP persistence rate	57.99 (10.41)	58.72 (10.81)	62.44 (9.22)	51.90 (7.40)
FTF overall CSU graduation rate	53.99 (11.16)	54.77 (10.53)	59.88 (9.62)	45.74 (9.44)
FTF overall CSU persistence rate	62.15 (8.70)	63.12 (8.67)	64.22 (8.65)	57.49 (7.12)
LD EOP graduation rate	68.51 (24.63)	66.57 (27.52)	82.04 (16.31)	60.62 (18.11)
LD EOP persistence rate	70.27 (24.57)	68.08 (27.79)	82.92 (16.50)	63.82 (17.75)
LD overall CSU graduation rate	74.08 (8.71)	73.93 (7.06)	81.68 (4.38)	66.34 (9.04)
LD overall CSU persistence rate	75.50 (8.18)	75.25 (6.68)	82.56 (4.22)	68.61 (8.74)

[†] Mean (SD)

Results

Result 1: Statewide EOP enrollment changed nonlinearly over time

Statewide EOP enrollment declined in the early 2000s, recovered through much of the 2010s, and then declined again after 2020. The year-to-year changes after 2020 were mostly negative, suggesting a possible disruption period rather than a steady long term linear trend. The initial exploration described the pattern as nonlinear, and high variations across campuses. The campuses with the greatest enrollment variability included San Diego, Fresno, San Francisco, Northridge, and Los Angeles. This supports the decision to include campus-level effects in the models, and that the repeated campus structure made mixed effects models appropriate .

Result 2: EOP graduation and persistence rates were lower than overall CSU rates, but the gap changed over time

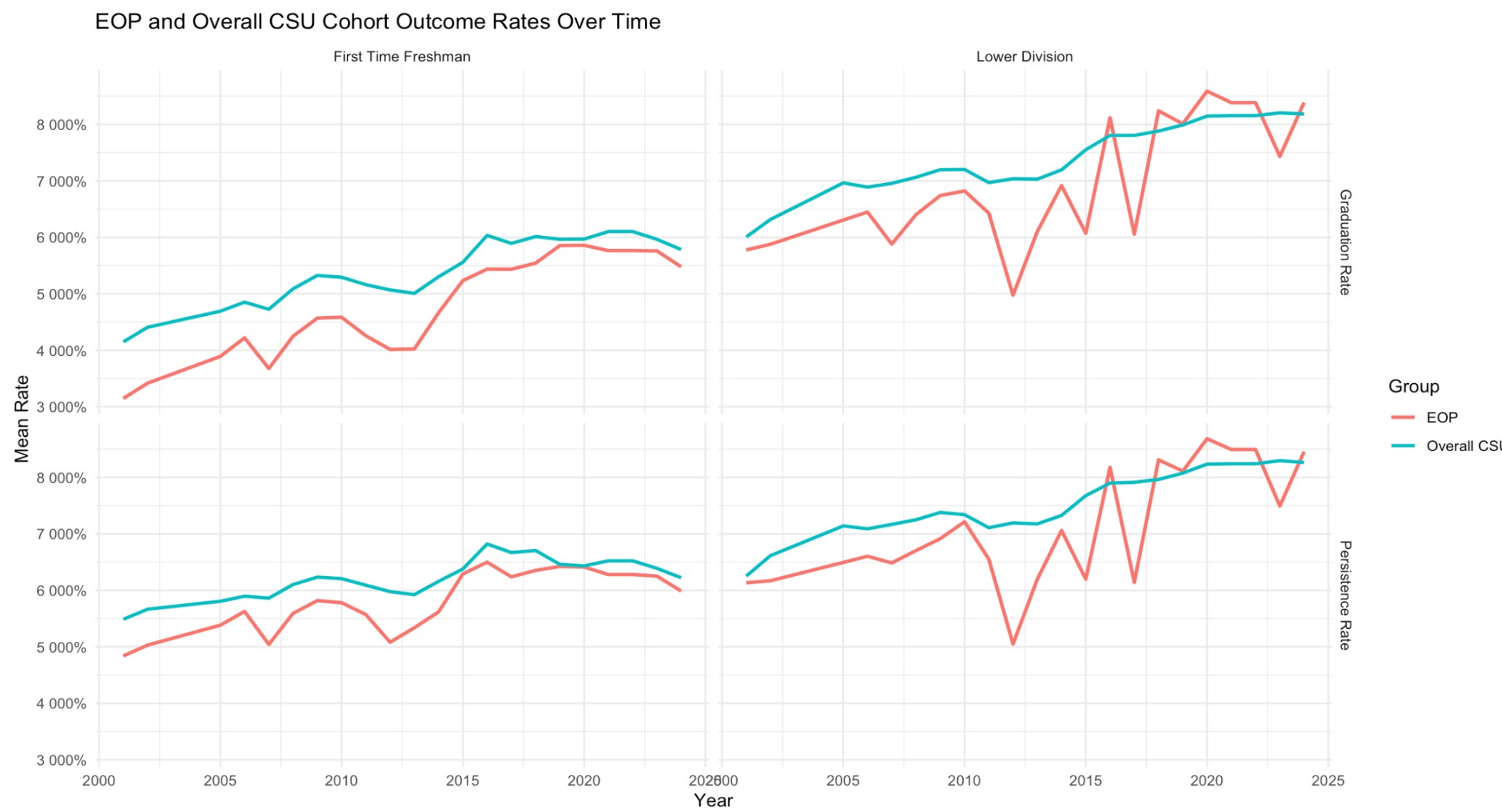
For first time freshman outcomes, EOP students had lower average rates than the overall CSU comparison group . A Wilcoxon signed rank test with a pvalue <0.001, and mixed effects model with negative coefficients supporting this finding.

Outcome Gap	EOP minus overall CSU
Graduation Rate Gap	-5.53 percentage points
Persistence Rate Gap	-3.60 percentage points

Table 7: Mixed-effects model comparing EOP with Overall CSU outcomes

effect	term	estimate	std.error	statistic
fixed	(Intercept)	-1189.356	62.344	-19.077
fixed	groupEOP	-620.477	89.489	-6.934
fixed	year	0.617	0.031	19.943
fixed	outcomePersistence Rate	9.240	0.293	31.483
fixed	groupEOP:year	0.306	0.044	6.882

However, the positive **groupEOP:year** interaction suggests that the EOP and overall CSU gap narrowed over time.



Results

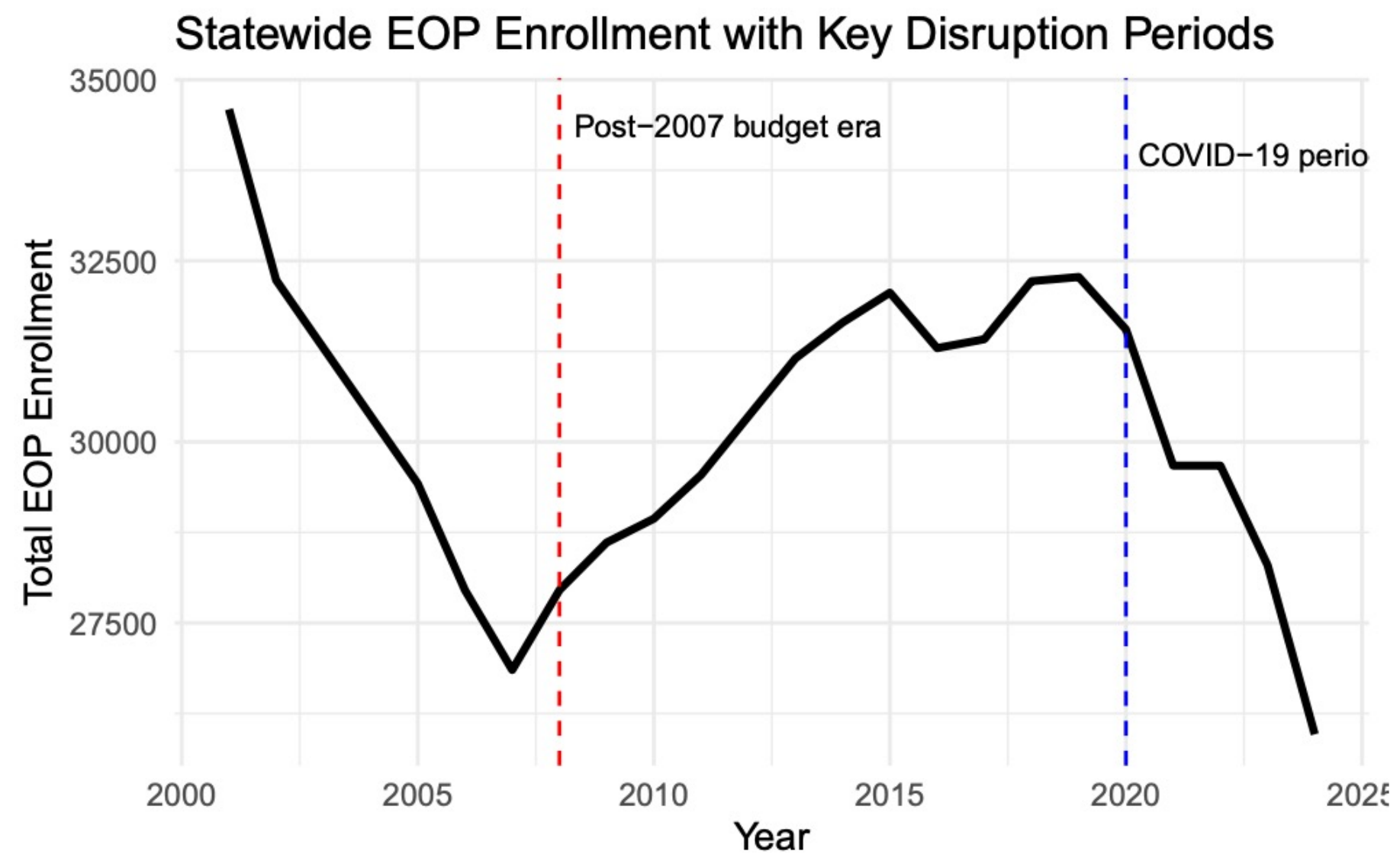
Result 3: Chico State generally followed the broader CSU pattern

Chico State EOP enrollment was **1,234 students in 2001**, remained near 1,100–1,200 students for much of the 2000s and 2010s, and declined to **954 students by 2024**. Chico’s enrollment pattern shows a gradual decline after 2020, similar to the broader CSU trend. This suggesting there are system level factors rather than only campus specific causes .

Result 4: Disruption Periods

The disruption period analysis supports the idea that 2008 and 2020 are useful breakpoints. The 2008 to 2019 period looks like a recovery period for enrollment, while the 2020+ period shows lower average enrollment and a negative slope change

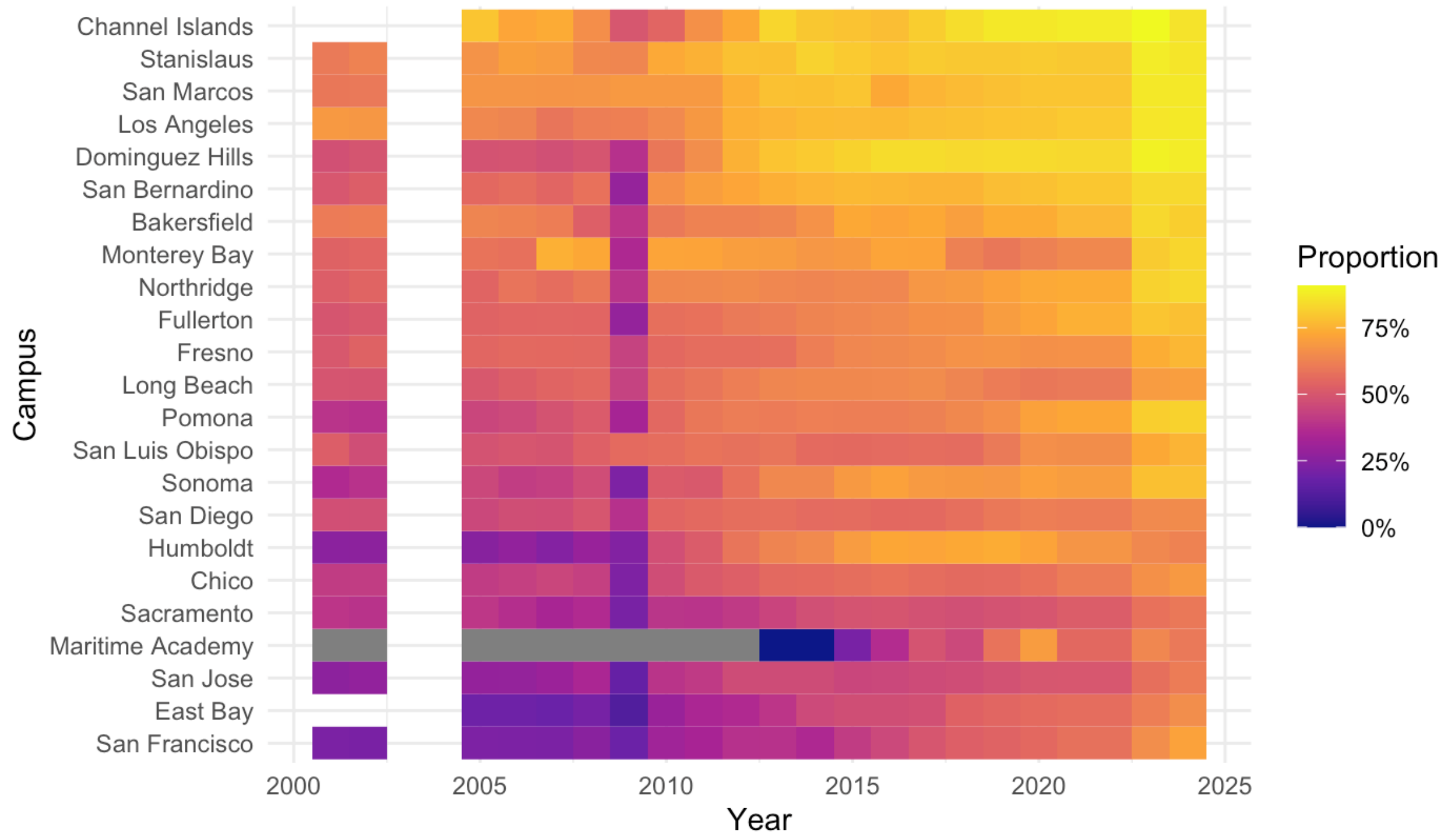
A key factor to these are the economical climate surrounding these times. The Education System was facing a series of budget cuts in the early 2000’s, resulting in a decline of over half of their funding with an estimation of \$81.1 million in 2000-01 to \$29.3 million in 2004-05.



Result 5: Demographics

The demographics show that Hispanic/Latino students make up the largest share of EOP enrollment and that this share increases over time. Some smaller categories are more variable because they have smaller counts and because reporting categories changed across years. The heat map shows that the Hispanic EOP share is not the same across campuses, with some campuses consistently having much higher proportions than others. The chi-square test of ($X^2=5916$) and (p-value < 2.2e-16) confirms that the overall distribution of ethnicity categories differs between 2001 and 2024, but it should be treated only as a snapshot because it ignores the repeated campus-year structure.

Heat Map of Hispanic EOP Share by Campus and Year



Conclusion

Conclusion

CSU EOP enrollment did not follow a simple straight line trend from 2001 to 2024. Instead, enrollment declined in the early years, recovered through much of the 2010s, and declined again after 2020. Campus differences were large, which means statewide averages alone do not fully describe EOP participation across the CSU system.

EOP student success rates were lower than the overall CSU comparison group for first time freshman graduation and persistence outcomes. However, the model results suggest that this gap changed over time. Chico State generally followed the broader CSU enrollment pattern, especially the decline after 2020, which suggests that system wide conditions may have influenced local campus trends.

Implications

These findings matter because EOP is connected to college access, retention, and graduation for students who may face educational and economic barriers. Declining enrollment after 2020 may indicate that EOP programs need continued monitoring and support, especially as campuses respond to changing enrollment patterns. The campus level differences also suggest that EOP outcomes may depend on local campus capacity, student population, and support structures.

For CSU leaders and campus programs, this analysis highlights the importance of tracking both enrollment and student success outcomes over time. A statewide average can hide important campus level differences, so future evaluation should consider campus specific trends when assessing EOP program needs.

Limitations

A few limitations should be kept in mind:

- Some missing values are structural because campuses or reporting categories were introduced later.
- The dataset supports overall EOP and total cohort outcomes, but it does **not** provide graduation or persistence rates separately by demographic group. Due to this, demographic analyses here focus on **composition of EOP enrollment**, not demographic specific graduation rates.

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